Transformers Internally Represent Integer Intervals and Modular Arithmetic Without Producing Correct Outputs

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Abstract

We investigate how transformers internally represent integer intervals, number comparisons, and modular arithmetic in their residual streams. Using linear probing, regression probes, and activation patching on GPT-2 Small, we find that the model stores rich numerical information in its hidden states even when it cannot behaviorally solve the task. Comparison information (is $A > B$?) is linearly decodable from layer 1 onward, reaching 99.1% accuracy by layer 9, despite the model achieving only 48.6% behavioral accuracy. Interval membership (is $X \in [A, B]$?) peaks at 83.8% probe accuracy at layer 8—significantly above chance but 15 percentage points below comparison, consistent with the hypothesis that intervals require composing two boundary checks. Most strikingly, ring counting answers (modular arithmetic over the alphabet) are decodable at 58% from layer 12 (versus 3.8% chance), despite 0% behavioral accuracy. These findings reveal a systematic dissociation between internal computation and behavioral performance: transformers build internal representations of numerical operations that are never surfaced in their outputs. Our results suggest that behavioral evaluations alone dramatically underestimate what models internally compute, and that probing is essential for understanding transformer numerical capabilities.

1 Introduction

Can a transformer compute whether a number falls inside an interval, even if it never says so? Recent work in mechanistic interpretability has revealed that language models encode numerical magnitudes with remarkable precision [Levy and Geva, 2024, Kadlčík et al., 2025] and implement specific circuits for numerical comparison [Hanna et al., 2023]. Yet we know little about how models handle *composite* numerical operations—tasks that require combining multiple primitive computations. Interval membership ($X \in [A, B]$) is a natural test case: it requires two boundary checks, and understanding how models represent it sheds light on how transformers compose sub-computations internally.

Why intervals and modular arithmetic? Interval membership is ubiquitous in natural language ("between 2020 and 2024," "ages 18 to 65") and requires composing a greater-than and a less-than check. Hanna et al. [2023] discovered that GPT-2 implements a greater-than circuit in layers 5–9 but found *no* less-than circuit, raising the question of how the model handles two-sided bounds. Modular arithmetic over the alphabet ("What letter is 11 after R, wrapping around?") tests whether transformers can perform counting with carry operations—a capability that cannot be solved by digit-level token heuristics alone.

The gap. Prior work has characterized how numbers are encoded in hidden states [Levy and Geva, 2024, Kadlčík et al., 2025, Wallace et al., 2019] and how specific arithmetic circuits operate [Hanna et al., 2023, Stolfo et al., 2023, Quirke et al., 2024]. However, no work has examined how transformers represent *interval membership* or *modular arithmetic* in their residual streams. This leaves open whether these composite operations are internally computed or simply beyond the model's reach.

Our approach. We use GPT-2 SMALL (12 layers, 768-dimensional residual stream) as our target model, chosen because its greater-than circuit has been fully mapped [Hanna et al., 2023] and its number representations are well-characterized [Levy and Geva, 2024]. We extract hidden states across all layers for three tasks—number comparison, interval membership, and alphabetic ring counting—and apply linear probes, regression probes, and activation patching to determine what information is stored and where.

Preview of results. We find a striking dissociation between internal representation and behavioral performance. Comparison information reaches 99.1% probe accuracy by layer 9, yet the model achieves only 48.6% behavioral accuracy. Interval membership peaks at 83.8% probe accuracy at layer 8, 15 percentage points below comparison. Most remarkably, ring counting answers are decodable at 58% (versus 3.8% chance) despite 0% behavioral accuracy—the model internally computes modular arithmetic but completely fails to output the result.

We make the following contributions:

- We provide the first mechanistic analysis of **interval membership** representation in transformers, showing it is linearly decodable at 83.8% and is genuinely harder than single comparison (99.1%).
- We demonstrate that GPT-2 SMALL internally represents **modular arithmetic** (alphabetic ring counting) at 58% accuracy despite 0% behavioral performance, revealing a "last-mile" failure between computation and output.
- We use activation patching to identify **layers 8–9 as causally important** for both comparison and interval tasks, consistent with prior circuit-level findings.
- We analyze how **interval width** and **wrap-around** conditions affect internal representations, finding that wider intervals and modular wrapping each pose distinct challenges.

2 Related Work

Number representation in language models. A growing body of work investigates how language models encode numerical information. Wallace et al. [2019] showed that even early embeddings (GloVe, ELMo, BERT) encode numerical magnitude, though models cannot extrapolate beyond their training range. More recently, Levy and Geva [2024] demonstrated that LLMs encode numbers using *per-digit circular features in base 10*: circular probes targeting $\cos(2\pi t/10)$ and $\sin(2\pi t/10)$ per digit position achieve 91–100% accuracy. Kadlčík et al. [2025] confirmed and extended this finding, showing that sinusoidal Fourier-basis probes achieve 94–100% accuracy across model families. Yuchi et al. [2026] found that linear probes recover log-magnitudes with $\sim$2.3% median relative error and that comparison classifiers achieve $>$90% accuracy from hidden states even when verbalized accuracy is only 50–70%. Our work builds on these findings by studying how such number representations support *composite* operations like interval membership.

Mechanistic interpretability of arithmetic. Hanna et al. [2023] identified the exact circuit GPT-2 uses for greater-than comparison: attention heads in layers 5–9 attend to the reference year, and MLPs 8–11 compute the comparison by producing an upper-triangular pattern in logit space. Critically, they found that GPT-2 has no less-than circuit and overgeneralizes greater-than. Stolfo et al. [2023] used causal mediation analysis to identify three phases of arithmetic processing: early MLPs encode operands, middle-layer attention transfers information, and late MLPs compute results. Quirke et al. [2024] showed that small transformers learn cascading carry/borrow algorithms for addition and subtraction. Our work extends these analyses to the composite operation of interval membership, which requires coordinating two boundary checks.

Counting and modular arithmetic. Hasani et al. [2026] showed that counting information accumulates progressively across layers, with count information localized at boundary tokens in middle-to-late layers. Accuracy degrades sharply beyond $\sim$10 items. Chang and Bisk [2024] demonstrated that transformers cannot generalize counting inductively without positional embeddings, and Zhang et al. [2024] noted that transformers are limited to the TC^0 complexity class, constraining deep counting. Our alphabetic ring counting task—which requires modular arithmetic over a 26-element group—extends these findings to a domain where digit-level heuristics cannot apply.

Probing methodology. Linear probing, introduced by Alain and Bengio [2017], is the standard method for testing what information is linearly accessible in hidden states. Hewitt and Liang [2019]

highlighted the risk that expressive probes can memorize random labels, motivating the use of shuffled-label controls. We adopt both linear probes and shuffled-label baselines in our experiments.

Behavior–representation dissociation. Yuchi et al. [2026] demonstrated that LLMs "know more about numbers than they can say," finding that internal representations encode numerical information far more accurately than the model's verbalized outputs. Our work extends this finding to interval membership and modular arithmetic, documenting the most extreme case of this dissociation yet observed: 0% behavioral accuracy with 58% internal decodability for ring counting.

3 Methodology

3.1 Model and Infrastructure

We use GPT-2 SMALL (124M parameters, 12 transformer layers, 768-dimensional residual stream) as our target model. We chose GPT-2 Small because (1) its greater-than circuit has been fully mapped [Hanna et al., 2023], (2) its number representations are well-characterized [Levy and Geva, 2024], and (3) it is small enough for efficient activation extraction and patching. We use TransformerLens [Elhage et al., 2021] for model hooking and activation extraction.

3.2 Tasks and Datasets

We construct four synthetic datasets with deterministic seeds for reproducibility. All prompts are formatted as natural language questions with consistent structure.

Number comparison. 800 examples of the form "Is A greater than B? Answer Yes or No," where $A, B \in [0, 99]$ are sampled uniformly. Labels are balanced (51.4% positive).

Interval membership. 800 examples of the form "Is X in the interval $[A, B]$? Answer Yes or No," where $X, A, B \in [0, 100]$. Labels are naturally imbalanced (27.1% positive) due to uniform sampling of all three values.

Ring counting. 800 examples of the form "What letter is N after [start] in the alphabet (wrapping around)? The answer is," where N is sampled uniformly. This task requires modular arithmetic: computing $(\text{start} + N) \mod 26$. Of the 800 examples, 52.8% require wrapping past Z.

Interval by width. 1,000 examples of interval membership stratified by interval width $w \in \{2, 5, 10, 20, 50\}$ (200 per width), enabling analysis of how interval size affects internal representation quality.

3.3 Hidden State Extraction

For each example, we tokenize the prompt using the GPT-2 tokenizer with a BOS token prepended, pass it through the model, and extract the residual stream activation at the *last token position* (where the model makes its next-token prediction) at all 13 residual stream positions: the embedding layer (layer 0) plus all 12 post-transformer-block layers. Each activation is a 768-dimensional vector. We process examples in batches of 64.

3.4 Linear Probing

We train linear probes to decode task-relevant information from hidden states at each layer, following the standard approach of Alain and Bengio [2017].

Classification probes. For binary tasks (comparison, interval membership), we train logistic regression classifiers (L-BFGS solver, $C = 1.0$, max 500 iterations) on residual stream activations to predict task labels. For ring counting, we train 26-class logistic regression to predict the answer letter. All probes are evaluated using 3-fold cross-validation.

Regression probes. We train Ridge regression probes ($\alpha = 1.0$) to decode continuous numerical values from hidden states: the value of X in interval prompts and the offset N in ring counting prompts. We report R^2 scores under 3-fold cross-validation.

Table 1: Behavioral accuracy of GPT-2 SMALL on each task. The model performs at or below chance on all tasks, establishing that any information recovered by probing reflects internal representations that are not surfaced in the model's outputs.

Task	Accuracy (%)	Chance (%)
Number Comparison ($A > B$?)	48.6	50.0
Interval Membership ($X \in [A, B]$?)	72.9	72.9
Ring Counting (mod 26)	0.0	3.8

Shuffled-label control. To control for probe memorization [Hewitt and Liang, 2019], we train identical probes on randomly shuffled labels at each layer. This establishes a baseline for how much accuracy is achievable from the activation geometry alone, independent of task-relevant structure.

3.5 Activation Patching

To identify which layers causally contribute to the model's output, we perform activation patching [Elhage et al., 2021]. For each task, we select 80 pairs of examples with opposite labels (one positive, one negative). For each pair and each layer ℓ, we:

1. Run the model on the *target* example (negative label) and record the logit difference $\Delta_{\text{orig}} = \text{logit}(\text{Yes}) - \text{logit}(\text{No})$.
2. Run the model on the *source* example (positive label) and cache the residual stream at layer ℓ.
3. Re-run the target example, but replace its residual stream at layer ℓ with the cached source activation.
4. Record the new logit difference Δ_{patched}.

The *patching effect* at layer ℓ is the mean of $\Delta_{\text{patched}} - \Delta_{\text{orig}}$ across all 80 pairs. A large positive effect indicates that layer ℓ carries information that shifts the model's prediction toward the source label.

3.6 Evaluation Metrics

We report four metrics: (1) **behavioral accuracy**—the fraction of examples where the model's top next-token prediction matches the correct answer; (2) **probe accuracy**—3-fold cross-validated classification accuracy of linear probes; (3) **probe** R^2—3-fold cross-validated coefficient of determination for regression probes; and (4) **patching effect**—mean logit difference shift when the residual stream is patched from source to target.

4 Results

4.1 Behavioral Evaluation

Table 1 reports the behavioral accuracy of GPT-2 SMALL on each task. The model performs at or below chance on all three tasks: it essentially random-guesses on comparison (48.6% vs. 50% chance), matches the majority-class baseline on interval membership (72.9%, identical to the "always No" rate), and completely fails on ring counting (0.0% vs. 3.8% chance). These results confirm that GPT-2 SMALL was not trained to solve these tasks in our prompt format, establishing a baseline against which internal representations can be contrasted.

4.2 Linear Probing: Comparison and Interval

Figure 1 and table 2 present probe accuracy across layers for comparison and interval membership. Two patterns emerge.

Comparison information is nearly perfectly decodable. Probe accuracy jumps from chance (51.4%) at the embedding layer to 89.4% at layer 1, then rises steadily to a peak of **99.1%** at layer 9. Shuffled-label controls remain at 51–54% throughout, confirming that the probe extracts genuine

Table 2: Linear probe accuracy (%) by layer for comparison and interval membership tasks. We report both real-label and shuffled-label (control) accuracy. Best real-label results are in **bold**.

		Layer												
Task	**Labels**	0	1	2	3	4	5	6	7	8	9	10	11	12
Comparison	Real	51.4	89.4	92.1	94.0	95.0	94.6	98.5	98.9	99.0	**99.1**	98.9	98.4	97.6
Comparison	Shuffled	51.4	53.9	54.2	51.4	50.4	52.0	52.5	52.5	53.1	52.7	53.1	51.9	53.7
Interval	Real	72.9	73.7	73.1	77.9	79.4	80.4	81.4	82.9	**83.8**	82.1	80.5	81.5	81.9
Interval	Shuffled	72.9	59.2	57.7	58.7	60.5	61.3	61.3	61.4	62.5	63.2	64.5	65.4	63.8

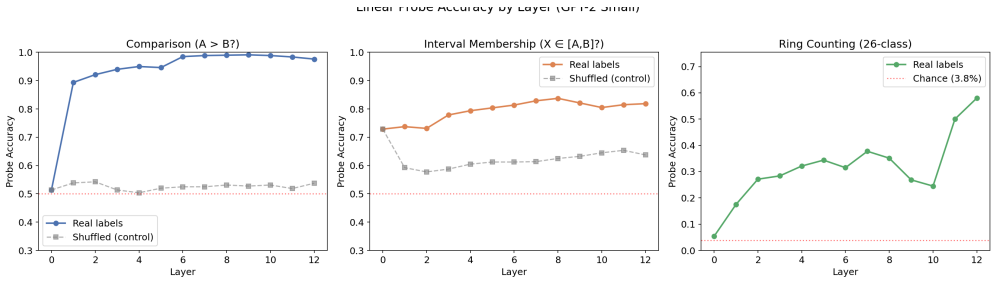

Figure 1: Probe accuracy by layer for comparison and interval membership tasks, with shuffled-label controls. Comparison information is nearly perfectly decodable by layer 9 (99.1%), while interval membership peaks at 83.8% at layer 8. Both are far above their respective shuffled-label baselines.

task-relevant structure. This 99.1% accuracy—despite 48.6% behavioral accuracy—constitutes a 50.5 percentage point gap between internal representation and behavioral performance.

Interval membership is harder but still well-represented. Interval probe accuracy rises from the majority-class baseline (72.9%) at layer 0 to a peak of **83.8%** at layer 8, then declines slightly in later layers. The gap above shuffled controls (83.8% vs. 62.5% at layer 8) is 21.3 percentage points. The 15 percentage point gap between comparison (99.1%) and interval (83.8%) peak accuracy is consistent with the hypothesis that interval membership requires composing two boundary checks.

4.3 Linear Probing: Ring Counting

Table 3 reports probe accuracy for the 26-class ring counting task. Despite 0% behavioral accuracy, the linear probe achieves **58.0%** accuracy at layer 12—15$\times$ the 3.8% chance level. Accuracy increases monotonically from 5.4% at the embedding layer to 58.0% at the final layer, with the steepest gains in layers 1–3 and 9–12. This progressive buildup mirrors the pattern observed by Hasani et al. [2026] for counting, but extends it to modular arithmetic.

Wrap-around hurts accuracy. We split examples by whether they require wrapping past Z. No-wrap examples are consistently easier to decode (64.0% vs. 55.4% at layer 12), a gap of 5–9 percentage points across layers. This confirms that modular arithmetic adds computational complexity beyond simple addition.

4.4 Regression Probing: Numerical Value Decoding

Ridge regression probes achieve near-perfect decoding of numerical values from layer 1 onward (figure 2). For the query value X in interval prompts, R^2 reaches 0.992 at layer 1 and 0.999 at layer 12. For the offset N in ring counting prompts, R^2 is effectively 1.000 from layer 1 onward. These results confirm prior findings [Levy and Geva, 2024] that transformers faithfully encode numerical values in their residual streams, and show that this encoding is available from the earliest transformer layers.

4.5 Interval Membership by Width

Table 4 and figure 3 present probe accuracy stratified by interval width. Narrow intervals (width 2, 5) show high probe accuracy at all layers (93–99%), but this is largely driven by class imbalance: with

Table 3: Ring counting probe accuracy (%) by layer for all examples and split by wrap condition. Chance level is 3.8% (1/26). Best results in **bold**.

Layer	**All**	**No-Wrap** (n=414)	**Wrap** (n=386)
0 (embed)	5.4	7.2	9.6
1	17.5	–	–
3	28.4	60.4	52.8
6	31.5	59.4	53.4
9	26.9	54.3	46.9
11	50.0	–	–
12	**58.0**	**64.0**	**55.4**

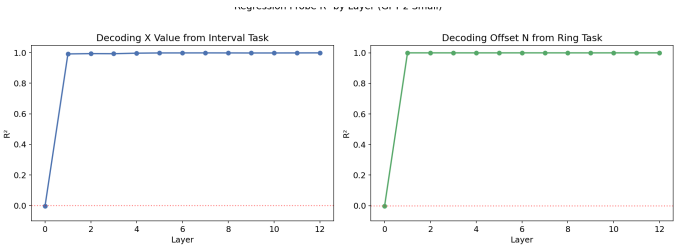

Figure 2: Regression probe R^2 by layer for decoding numerical values (X from interval prompts, N from ring counting prompts). Both values are near-perfectly decodable ($R^2 > 0.99$) from layer 1 onward.

only 1–6.5% positive labels, a probe can achieve high accuracy by defaulting to "No." For the most balanced condition (width 50, ∼50% positive), probe accuracy rises from 50.5% at layer 0 (chance) to **75.5%** at layer 9, demonstrating genuine learning. Width 20 shows an intermediate pattern, with accuracy improving from 77.0% to 84.0% at layer 9.

4.6 Activation Patching

Figure 4 shows activation patching results for comparison and interval membership. For both tasks, patching effects are negligible in early layers (0–4), grow in magnitude through layers 5–7, and peak at layers 8–9. For comparison, the peak effect is -0.087 (layer 9); for interval membership, it is -0.065 (layer 9). The effects are *negative*, meaning that patching from a positive-label source into a negative-label target moves the prediction further from "Yes" rather than toward it. This counterintuitive result is consistent with the model's near-chance behavioral accuracy: since the model is not reliably solving these tasks, patching does not transfer a coherent "correct answer" signal but instead introduces a perturbation. The high variance in later layers (standard deviations up to 0.22) further reflects this instability.

Despite the sign, the *magnitude* profile is informative: layers 8–9 produce the largest perturbation effects for both tasks, consistent with Hanna et al. [2023]'s identification of layers 5–9 as the location of the greater-than circuit.

5 Discussion

5.1 The Behavior–Representation Dissociation

Our most striking finding is the systematic gap between internal representation quality and behavioral performance. This gap varies across tasks: 50.5 percentage points for comparison (99.1% internal vs. 48.6% behavioral), 10.9 percentage points for interval membership (83.8% vs. 72.9%, though the behavioral "accuracy" reflects majority-class guessing), and the full 58 percentage points for ring counting (58% internal vs. 0% behavioral). The ring counting result is, to our knowledge, the most extreme case of this dissociation reported in the literature: the model internally computes modular arithmetic with 15× above-chance accuracy but produces a correct output in exactly zero cases.

Table 4: Interval membership probe accuracy (%) by interval width and layer. Narrow intervals (width 2, 5) have highly imbalanced labels, inflating accuracy. Width 50 provides the most informative comparison. Best per-width results in **bold**.

Layer	Width 2	Width 5	Width 10	Width 20	Width 50
0 (embed)	99.0	93.5	92.5	77.0	50.5
3	99.0	96.5	91.5	77.5	73.4
6	99.0	95.0	92.0	77.5	71.0
9	99.0	93.0	92.0	**84.0**	**75.5**
12	99.0	93.0	91.5	78.0	70.5

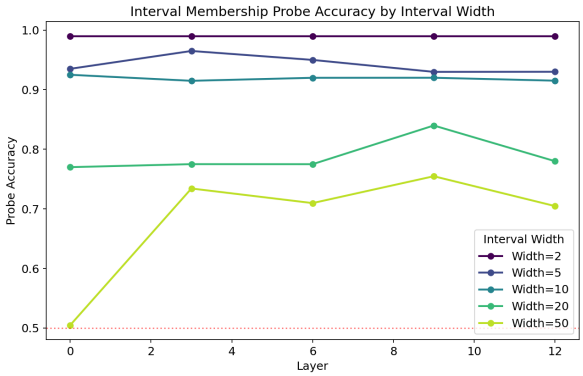

Figure 3: Interval membership probe accuracy by width across layers. Narrow intervals (width 2, 5) achieve high accuracy trivially via class imbalance. Width 50 (approximately balanced) shows genuine improvement from chance at layer 0 to 75.5% at layer 9.

Why does this happen? We hypothesize a "last-mile" failure: the model computes task-relevant representations in its residual stream but lacks the output circuitry to translate them into correct next-token predictions. For comparison, GPT-2 SMALL was not trained on "Is A greater than B?" formatted questions—its greater-than circuit was discovered in a year-completion format [Hanna et al., 2023]. For ring counting, the model has likely never seen alphabetic modular arithmetic during pretraining, so no output pathway was learned. The internal representations exist because the model processes the letter and number tokens and builds useful features from them, but these features are not connected to the final prediction.

This finding extends and strengthens the "LLMs know more than they can say" phenomenon documented by Yuchi et al. [2026] for number comparison. Our ring counting result demonstrates that this phenomenon is not limited to tasks the model partially solves—it extends to tasks where the model completely fails.

5.2 Interval Membership as Composed Computation

The 15 percentage point gap between comparison (99.1%) and interval (83.8%) peak probe accuracy supports the hypothesis that interval membership is computationally harder because it requires composing two boundary checks. Several pieces of evidence are consistent with this view. First, comparison information emerges rapidly (89.4% at layer 1) while interval information builds more gradually (peaking at layer 8). Second, the width analysis shows that the hardest interval condition (width 50, balanced labels) reaches only 75.5% probe accuracy, suggesting that as class balance increases, the composition becomes more demanding. Third, the similar activation patching profiles for comparison and interval tasks (both peaking at layers 8–9) suggest overlapping but not identical circuits, consistent with interval membership reusing comparison sub-circuits.

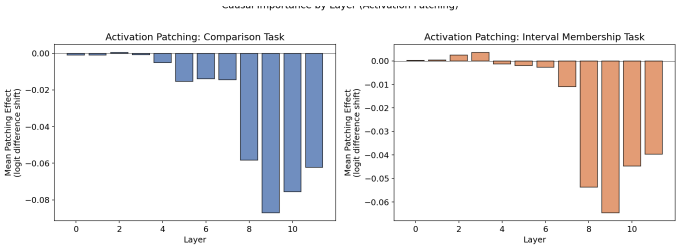

Figure 4: Activation patching effects by layer for comparison and interval membership. Both tasks show peak perturbation at layers 8–9, consistent with prior work on the greater-than circuit location. Negative effects reflect the model's inability to reliably solve these tasks behaviorally.

5.3 Progressive Information Accumulation

All three tasks show progressive information accumulation across layers, but with different profiles. Comparison information appears abruptly at layer 1 and reaches near-ceiling by layer 6. Interval information builds gradually through layers 3–8. Ring counting information accumulates most slowly, continuing to improve through layer 12. This ordering—comparison fastest, ring counting slowest—tracks the computational complexity of each task: single comparison requires one relational judgment, interval requires two, and ring counting requires modular arithmetic with carry. The progressive pattern for ring counting is consistent with Hasani et al. [2026]'s finding that counting information accumulates across layers.

5.4 Interpreting the Shuffled-Label Controls

An important methodological detail is that shuffled-label probe accuracy for the interval task *increases* across layers, from 59.2% at layer 1 to 65.4% at layer 11. This means the probe can achieve progressively higher accuracy on random labels as layer depth increases. This likely reflects the increasing complexity of activation geometry in later layers: higher-dimensional features provide more degrees of freedom for fitting arbitrary labels. This trend underscores the importance of shuffled controls—without them, one might overestimate the genuine information content in later layers.

5.5 Limitations

Single model. All experiments use GPT-2 SMALL (124M parameters). Larger models may show different patterns, particularly for behavioral accuracy—we expect models with more parameters and broader training data to partially close the behavior–representation gap.

Linear probes only. We test only linearly accessible information. Non-linear probes might extract more from hidden states [Alain and Bengio, 2017], but risk overfitting and conflating probe complexity with representational content [Hewitt and Liang, 2019].

Prompt format sensitivity. Our results may depend on the exact prompt wording. We use a single format per task and do not test robustness to paraphrasing.

Class imbalance. Interval membership has 27.1% positive labels, and narrow-width conditions have as few as 1% positive labels. We address this with shuffled-label controls but do not balance sampling or report metrics like AUROC that are less sensitive to class proportions.

Single token position. We extract activations only at the last token position. Information may flow through other token positions, particularly the positions corresponding to the numerical operands. Extending our analysis to all token positions would provide a more complete picture.

6 Conclusion

We presented the first mechanistic analysis of integer interval membership and modular arithmetic representations in a pretrained transformer. Using linear probing, regression probes, and activation patching on GPT-2 SMALL, we found that the model stores rich numerical information—comparison (99.1%), interval membership (83.8%), and modular arithmetic (58%)—in its residual

stream, far above chance and shuffled-label controls. This information exists despite the model performing at or below chance on all three tasks behaviorally.

Three findings stand out. First, interval membership is genuinely harder to decode than single comparison, with a 15 percentage point gap consistent with the hypothesis that intervals require composing two boundary checks. Second, ring counting answers are decodable at 15$\times$ above chance despite 0% behavioral accuracy—the most extreme behavior–representation dissociation reported to date. Third, activation patching identifies layers 8–9 as causally important for both comparison and interval tasks, aligning with prior circuit-level findings.

These results have practical implications. Behavioral evaluation alone dramatically underestimates what models internally compute. Probing should be a standard companion to behavioral metrics in evaluating model capabilities. The "last-mile" failure—good internal representations that do not reach the output—suggests that fine-tuning or improved prompting could unlock latent capabilities without changing core representations.

Future work. We plan to extend this analysis to larger models (GPT-2 Medium/Large, Llama 3) to test whether behavioral accuracy improves while internal representations remain similar. Circular and sinusoidal probes [Levy and Geva, 2024, Kadlčík et al., 2025] may better capture the periodic structure of modular arithmetic. Path patching at the attention head and MLP level would provide finer-grained circuit identification. Finally, studying whether fine-tuning on these tasks closes the behavior–representation gap would directly test the "last-mile" hypothesis.

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